
UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have advanced rapidly across the full spectrum of visual question answering—visual reasoning, document understanding, and multimodal dialogue alike—and these gains hold consistently across VLMs built on diverse base language models, alignment architectures, and training corpora. This very progress, however, obscures a systematic blind spot: on traditional image classification benchmarks, precisely the setting that probes the fine-grained visual knowledge on which vision-centric competence rests, these same models lag markedly. We resolve this disconnect through a comprehensive, unified evaluation of a large and representative set of recent VLMs on fine-grained classification, and, in doing so, isolate the factors that separate fine-grained knowledge from performance on the reasoning-oriented tasks that current benchmarks reward. A systematic program of controlled ablation experiments then establishes a clean decomposition of the contributing factors: a stronger base LLM raises every benchmark score in equal measure, whereas a stronger vision encoder confers a gain concentrated disproportionately on fine-grained classification. We further establish that the pretraining stage is the single decisive determinant of fine-grained performance—and decisively more so when the language-model weights are unfrozen during pretraining, so that the model can assimilate the visual signal rather than merely align to it. Taken together, these findings resolve the observed disconnect into a set of identifiable design levers and chart a concrete, actionable path toward VLMs with substantially stronger fine-grained visual understanding and, more broadly, more reliable vision-centric capability.

1 INTRODUCTION

Vision-language models (VLMs), which couple a vision encoder with a large language model (LLM), have rapidly acquired increasingly sophisticated capabilities for interpreting and reasoning about visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b), and now discharge demanding tasks that span open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). This trajectory of progress, however, has left a foundational question both unresolved and largely unexamined: how well do these models perform fine-grained visual perception, and—more consequentially—what concretely can be done to improve it? We confront that question directly, treating fine-grained recognition not as a peripheral curiosity but as a core axis of visual intelligence and a prerequisite for the very vision-centric competence that downstream reasoning presupposes.

Fine-grained visual recognition—the ability to discriminate among visually similar categories—requires a model to resolve subtle discriminative cues while remaining invariant to intra-class variation in pose, illumination, and background. Traditional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) attain strong fine-grained classification performance; the fine-grained behavior of the newer generation of VLMs—models constructed atop these very encoders—has, strikingly, gone comparatively unexamined, leaving open the question of how much of that encoder-resident capability survives multimodal integration.

Characterizing the fine-grained knowledge capabilities of VLMs is essential precisely because perception constitutes the foundation on which higher-order understanding and reasoning are erected: a

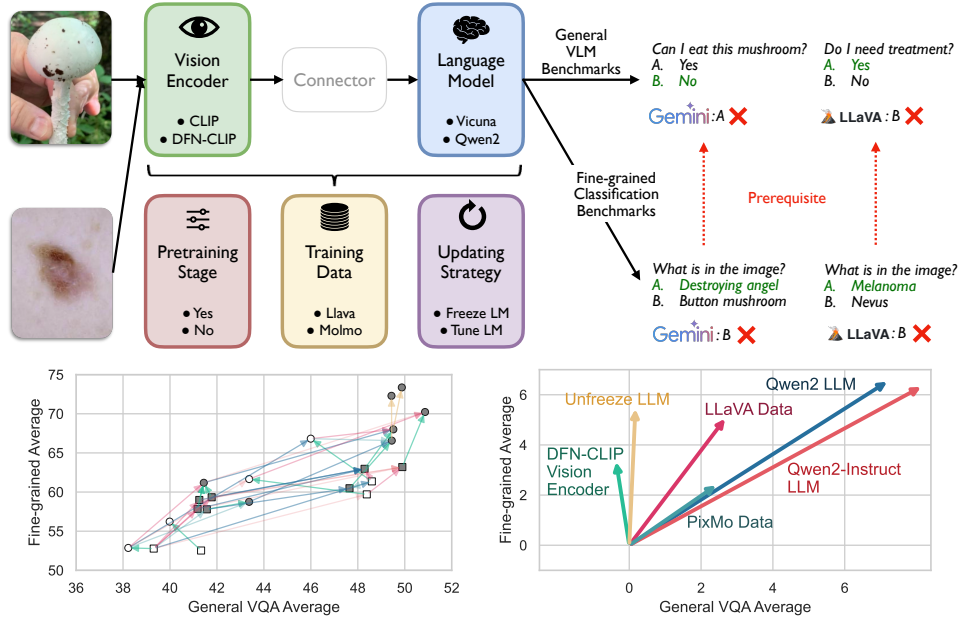


Figure 1: **Overview.** (Top) We investigate the fine-grained classification capabilities of vision-language models (VLMs)—a foundational yet routinely overlooked dimension of visual competence on which higher-level understanding and reasoning depend. (Bottom) Across 22 controlled ablation experiments that systematically vary the principal model components and training strategies (left), we isolate the factors that decisively govern fine-grained classification performance in VLMs (right), attributing the observed variation to a small set of identifiable design choices.

model cannot reason reliably about what it cannot first recognize. Real-world deployments of VLMs routinely confront fine-grained, long-tailed distributions in which discriminating among closely related categories is decisive rather than incidental. Should a model misidentify a mushroom species, for instance, no quantity of downstream textual knowledge can recover whether the specimen is poisonous or safe to consume. Beyond such immediate stakes, a principled account of the factors that govern fine-grained knowledge acquisition translates directly into design guidance for the architectures and training strategies of the next generation of VLMs—the actionable payoff this work is organized to deliver.

We first present a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs, directly juxtaposing their fine-grained classification performance against their capabilities on general VLM tasks. Concretely, we evaluate 15 representative VLMs—spanning LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024)—on four fine-grained classification benchmarks that jointly span the object, flower, pet, and food domains: ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014), and set these against aggregated results from eight general VLM benchmarks including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). This analysis establishes two facts. First, models of comparable general ability diverge substantially in fine-grained classification accuracy, demonstrating that fine-grained recognition constitutes a distinct axis of visual intelligence that current VLM benchmarks fail to capture and that general-purpose evaluation therefore systematically overlooks. Second, it exposes a persistent, systematic gap between VLMs and the CLIP-based vision encoders on which they are built—a gap that corroborates and materially extends prior observations (Zhang et al., 2024; Geigle et al., 2024) to a far broader and more representative model set, and thereby quantifies the fine-grained capability that current integration pipelines leave unrealized.

To pinpoint what actually drives fine-grained classification performance in VLMs, we then conduct a systematic study that disentangles the contributions of the principal model components and training paradigms. Concretely, we quantify how the two core architectural components—the language model and the vision encoder—together with the training strategy—the pretraining stage, the choice of which weights are updated, and the quality of the training data—each shape fine-grained classifi-

cation. Through 22 controlled ablation experiments, each holding the surrounding pipeline fixed so that it isolates a single factor, we identify the levers that decisively govern fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Upgrading the base language model lifts performance uniformly, delivering comparable gains on fine-grained classification and general VLM benchmarks alike; the LLM thus acts as a shared, non-selective driver of both capabilities.
- **Vision encoder (§4.2):** A stronger vision encoder selectively improves fine-grained classification while exerting only limited influence on general VLM benchmarks—an asymmetry that becomes most pronounced under the standard two-stage training regime and that pinpoints the encoder as a fine-grained-specific lever.
- **Pretraining stage (§4.3):** Large-scale pretraining on image-captioning data is the single most effective lever for fine-grained classification, conferring substantial gains while leaving general VLM benchmarks comparatively unaffected—identifying pretraining, rather than instruction tuning, as the locus of fine-grained knowledge acquisition.
- **Weight updating method (§4.3):** Jointly updating the vision-encoder-to-LLM connector and the LLM during pretraining improves fine-grained classification well beyond what connector-only pretraining attains, establishing which weights are updated—not merely whether pretraining occurs—as a decisive design choice.
- **Data quality (§4.3):** The quality of the pretraining and instruction-tuning data exerts only a limited effect on fine-grained classification—an effect that all but vanishes when the language-model weights are held frozen during pretraining, precisely locating the condition under which caption quality does and does not matter.

In sum, this study places fine-grained visual perception in VLMs—an essential capability that current benchmark evaluations systematically fail to surface—at the center of analysis, and, through controlled ablations, decomposes it into the specific architectural and training factors that govern its acquisition. These findings together constitute actionable guidance for building vision-centric VLMs with markedly stronger visual understanding, and thereby for improving their reliability across the real-world, long-tailed tasks that hinge on fine-grained recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) integrate visual inputs with a language model and predominantly instantiate the LLaVA-style architecture (Liu et al., 2023), which comprises three core components: a vision encoder, a language model, and an MLP connector that projects visual features into the language model’s input space. Such models attain strong performance across a broad range of multimodal tasks, encompassing visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). Their evaluation rests, correspondingly, on a large family of benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks, however, predominantly probe reasoning and language understanding conditioned on visual input, and in doing so leave the core vision-centric competencies—object recognition and fine-grained classification foremost among them—essentially unmeasured. We close this gap by systematically evaluating and analyzing VLMs on fine-grained classification, an essential yet underexplored dimension of visual intelligence, and treat it as a first-class evaluation axis rather than a corollary of reasoning ability.

Fine-grained visual classification. Fine-grained classification is a well-established computer vision task that seeks to discriminate among visually similar subcategories within a broader object category, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). The corresponding benchmarks have long served as central testbeds for training and evaluating vision encoders, ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021) among them. Despite this foundational role in vision research, recent VLMs have largely omitted these datasets from their evaluation protocols in favor of visual question answering, so that a capability central to the encoders they inherit goes unassessed in the integrated models. Because fine-grained classification is indispensable to real-world applications that involve long-tail distributions and demand precise recognition—medical diagnosis, food safety, and species identification among them—evaluating VLMs on such benchmarks is not optional but essential for

assessing their practical usability from a vision-centric standpoint, rather than on language-centric reasoning tasks alone.

Evaluation and ablation of VLMs for fine-grained classification. Prior studies of VLM evaluation on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024) remain limited in scope, typically confined to a handful of models or datasets, and stop short of explaining the performance they report. We substantially broaden this line of work along two axes: in coverage, evaluating a considerably wider and more representative range of VLMs across multiple fine-grained classification benchmarks within a single, unified multiple-choice protocol; and in depth, contributing a comprehensive controlled ablation that isolates the factors driving fine-grained classification performance—an explanatory analysis without precedent in this setting.

A parallel body of work examines the design space of VLMs more generally, dissecting model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of multimodal datasets (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). These studies, however, are oriented almost exclusively toward general benchmarks that emphasize language and reasoning, and their conclusions therefore need not transfer to the vision-centric regime. We instead trace how the central design choices—base model selection, pretraining methodology, and the composition of the pretraining and instruction-tuning data—shape fine-grained classification in particular, yielding novel and directly actionable design guidance for optimizing VLMs on vision-centric tasks.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section motivates the evaluation of fine-grained classification, specifies our benchmarks and evaluation protocol, and reports the central observations that emerge from testing 15 existing VLMs under a single unified protocol. Two findings anchor the analysis: fine-grained classification is a distinct, unmeasured axis of visual capability, and current VLMs recover only part of the fine-grained knowledge already present in their own vision encoders.

3.1 MOTIVATION

Fine-grained classification is a foundational computer vision task that targets the discrimination of visually similar subcategories within a broader category—distinct bird species or car models, for instance (Wei et al., 2021). A substantial suite of evaluation benchmarks, ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) chief among them, has become the standard testbed for training and evaluating vision encoders. With the advent of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) attain near- or super-human accuracy even on the most demanding of these fine-grained tasks, establishing that the representational capacity for fine-grained discrimination is by now well within reach of the encoders VLMs are built upon.

These benchmarks, however, have been conspicuously absent from the evaluation of vision-language models (VLMs). Although recent VLMs such as LLaVA (Liu et al., 2023) demonstrate strong capabilities in interpreting visual concepts, their evaluation remains largely confined to visual question answering (VQA)—academic VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Because these benchmarks foreground reasoning and language processing yet contain no fine-grained object categories, they leave the core vision-centric capabilities of VLMs essentially unexamined—a gap in the evaluation regime, not merely in the reported numbers.

Fine-grained classification is, nonetheless, decisive for VLMs, precisely because real-world applications routinely encounter long-tailed distributions that demand precise visual discrimination among closely related categories. Consider a blind user asking whether a mushroom is safe to eat: the model must correctly identify the species before it can reason about toxicity at all, and misclassifying a deadly destroying angel as an edible white button mushroom could prove fatal. The same logic governs medical diagnosis, where a failure to distinguish visually similar conditions can precipitate incorrect treatment and serious harm, and autonomous driving, where confusing a “stop” sign with a “do not enter” sign—both red and similarly shaped—can produce dangerous navigation errors.

Across these settings the pattern is identical: recognition is upstream of every subsequent inference, so excelling at fine-grained classification is not a peripheral desideratum but a prerequisite for the safety and reliability of VLMs in real-world deployment.

3.2 BENCHMARKS

We probe the fine-grained visual knowledge of VLMs with four well-established object recognition benchmarks, deliberately selected so that each targets a distinct domain of fine-grained classification and the four jointly span generic objects, flowers, pets, and food: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014). This spread ensures that our conclusions reflect fine-grained recognition as a general competence rather than an artifact of any single domain.

Because these benchmarks span 37 to 1,000 distinct classes, evaluating VLMs directly in the multiple-choice format standard for VLM training and evaluation is not straightforward: the full label set cannot be presented as answer options. We resolve this by adopting the methodology of Geigle et al. (2024) to recast each dataset as a set of 5-way multiple-choice questions. Concretely, for every image we employ an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) to retrieve, by image-label cosine similarity, the four hardest incorrect labels as negative choices, and combine them with the ground-truth label to form the five-way answer set. The choice of the OpenCLIP retrieval model is deliberate and controls a specific confound: because it is trained on a corpus distinct from that of OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), and DFN-CLIP (Fang et al., 2023), it prevents distractor difficulty from being systematically biased for or against any VLM that adopts one of those encoders. This construction lets us evaluate VLMs in their native multiple-choice format while preserving the bulk of the task’s intrinsic difficulty: CLIP ViT-L/14-336px (Radford et al., 2021), for instance, attains 75.2% accuracy on the full 102-class Flowers-102 and rises to only 78.3% on our 5-way version—a margin of just 3.1pp that confirms the reformulation retains the greater part of the discriminative challenge rather than trivializing it.

3.3 TESTING SETUP

We evaluate 15 recent VLMs spanning the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abdin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b), a set chosen to represent the principal contemporary model families rather than a single lineage. For each VLM we adopt its respective default prompt format from VLMEvalKit (Duan et al., 2024) and score accuracy by exact match against the multiple-choice options, so that every model is queried in the convention under which it was trained. As a reference upper bound, we evaluate a range of CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) under the standard zero-shot protocol, in which the image and each candidate class label (“A photo of a [class]”) are mapped through CLIP’s image and text encoders respectively and the top-1 prediction is taken as the label of maximal image–text cosine similarity.

To determine whether existing VLM benchmarks already capture fine-grained classification ability, we compare each VLM’s fine-grained accuracy against its average score across eight general VQA benchmarks from VLMEvalKit, MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024) among them; a systematic decoupling between the two would indicate that the general benchmarks do not subsume fine-grained competence.

3.4 RESULTS

Figure 2 plots fine-grained classification performance (averaged across our four benchmarks) against general VQA performance (averaged across eight existing VQA benchmarks) for every tested VLM. The two quantities decouple sharply: models with essentially identical general VQA performance span a wide range of fine-grained classification accuracy, so that a model’s standing on the general benchmarks predicts little about its fine-grained competence. The comparison between CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) makes this concrete: the two are matched almost exactly on general VQA—both reach nearly 48% average accuracy—yet their fine-grained classification accuracy differs by a full 19 percentage points (pp), 77.3% for

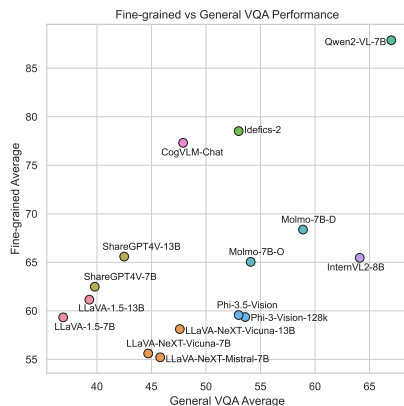


Figure 2: **Fine-grained classification versus general VQA performance across VLMs.** Each point is one of the 15 evaluated VLMs, plotting mean fine-grained classification accuracy (over our four benchmarks) against mean general VQA accuracy (over eight benchmarks). Fine-grained accuracy varies widely at essentially fixed general VQA performance, establishing fine-grained classification as a distinct axis of visual capability that standard VQA benchmarks do not measure.

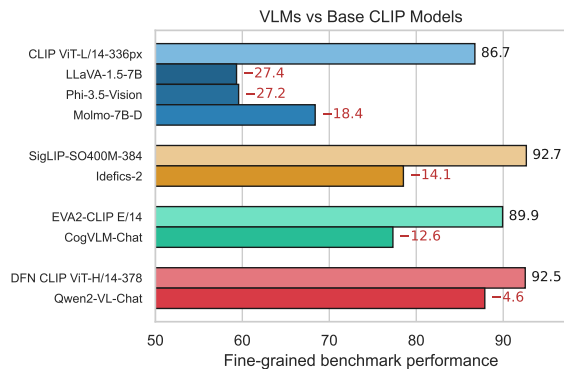


Figure 3: **VLMs versus their own CLIP vision encoders on fine-grained classification.** For each VLM we compare its fine-grained accuracy against that of the very CLIP encoder it is built upon, under the same 5-way protocol. Although Qwen2-VL-Chat nearly recovers the fine-grained accuracy of its DFN-CLIP vision encoder, every other VLM trails its own encoder by a wide margin, quantifying the fine-grained capability that current integration pipelines leave unrealized—and thereby the headroom our ablation analysis is designed to recover.

CogVLM-Chat against 58.1% for LLaVA-NeXT-Vicuna-13B. A discrepancy of this magnitude between models matched on general VQA demonstrates conclusively that fine-grained classification ability is not captured by existing VLM benchmarks, and establishes it as a distinct dimension of visual intelligence that must be measured in its own right.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Every current VLM is constructed on top of a pretrained vision encoder, and our multiple-choice reformulation turns this fact into a direct, like-for-like comparison between each VLM and the CLIP model it inherits on traditional classification benchmarks. Figure 3 contrasts the fine-grained classification accuracy of selected VLMs against that of their corresponding CLIP baselines, exposing a substantial and consistent shortfall rather than an isolated failure. The gap is starkest for Molmo (Deitke et al., 2024), a very recent model that excels across VQA benchmarks yet attains only 68.4% average accuracy in our multiple-choice setting, fully 18.3pp below the 86.7% reached by its own CLIP ViT-L/14 (Radford et al., 2021) vision encoder. Even the strongest model we test, Qwen2-VL-Chat (Wang et al., 2024b), trails its own state-of-the-art DFN-CLIP (Fang et al., 2023) vision encoder by 4.6 points—demonstrating that the shortfall persists even at the frontier, though it can evidently be narrowed. That fine-grained knowledge already resident in the vision encoder is not fully preserved through multimodal integration is thus the crux of the problem: the capability is present in the components but attenuated in the assembled model. This diagnosis directly motivates a systematic analysis of VLM design choices aimed at identifying the factors that recover and enhance fine-grained classification, which we undertake in the next section.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

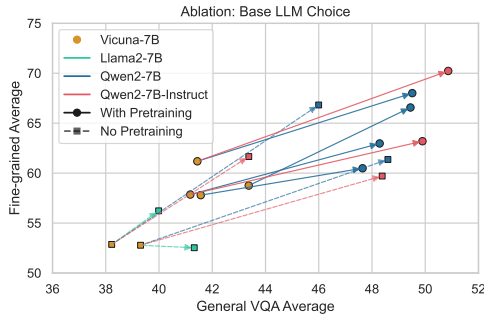


Figure 4: **Base-LLM ablation: replacing LLaVA’s Vicuna-7B with alternative LLMs.** Averaged across vision encoders and pretraining configurations, switching from Vicuna to Qwen2-7B raises fine-grained performance by +7.5pp and general VQA performance by +7.7pp—near-identical margins that identify the base LLM as a shared driver lifting both axes in tandem rather than a fine-grained-specific lever.

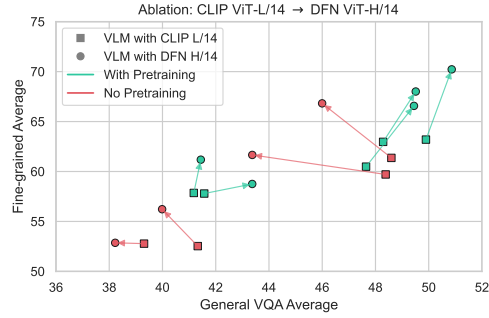


Figure 5: **Base vision-encoder ablation: replacing LLaVA’s CLIP ViT-L/14 with DFN-CLIP ViT-H/14.** Given sufficient pretraining, the stronger DFN-CLIP encoder improves fine-grained performance by +4.5pp while lifting general performance by only +1.2pp. The near-fourfold asymmetry between the two gains isolates a vision-encoder effect that is specific to—and disproportionately concentrated in—fine-grained classification, in direct contrast to the balanced LLM effect of Figure 4.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational results of Section 3, we now ablate the model-level differences that drive fine-grained classification performance, concentrating on the two dimensions the observational analysis implicates as most consequential: base model selection and training methodology. All experiments are conducted within the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023), and we hold the surrounding pipeline fixed so that each ablation varies a single factor in isolation and its effect can be read off unambiguously. Every model undergoes one epoch of pretraining (where applicable) followed by one epoch of finetuning, and we retain LLaVA-1.5’s default hyperparameters—learning rate and batch size included—throughout, so that no confound other than the ablated variable enters any comparison.

We report fine-grained performance as the mean accuracy across the four classification benchmarks introduced in Section 3.2, and general VQA performance as the mean over three widely used multiple-choice benchmarks—MMM (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b)—selected for their broad domain coverage and established role in VLM evaluation. Reporting both axes for every configuration lets each ablation reveal not only whether a factor helps, but whether its benefit is specific to fine-grained recognition or shared with general competence.

4.2 BASE MODELS

The analysis in Section 3 reveals that recent models have steadily adopted newer base LLMs while, in most cases, retaining the CLIP ViT-L/14 vision encoder of Radford et al. (2021)—an asymmetry in how the field has evolved its two core components. We therefore isolate how each of these two base-model choices—the language model and the vision encoder—individually shapes fine-grained classification performance, so that the contribution of each can be attributed rather than conflated.

4.2.1 LLM CHOICE

Setup. Taking LLaVA-1.5-7B (Liu et al., 2023) as our baseline, we vary the LLM component while sweeping across vision encoders and pretraining configurations, so that the LLM effect is estimated

as an average over—and hence independently of—the other design axes. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) used in LLaVA against Qwen2-7B (Yang et al., 2024) (the LLM of Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)), together with Llama-2-7B (Touvron et al., 2023) (Vicuna’s base model) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B), spanning both the base and instruction-tuned regimes.

Findings. As Figure 4 shows, replacing Vicuna-7B with Qwen2-7B improves performance substantially and uniformly across every configuration, with average gains of +7.5 percentage points (pp) on fine-grained benchmarks and +7.4pp on general VQA benchmarks—a symmetry that is itself the central result: the stronger LLM raises both axes essentially equally. The instruction-tuned Qwen2-7B-Instruct yields closely comparable gains (+7.5pp and +8.1pp, respectively), confirming the effect is robust to instruction tuning, whereas Llama2-7B, the non-instruction-tuned base of Vicuna, produces only modest improvements whose downstream implications, absent instruction tuning, remain difficult to interpret. Crucially, all of our trained models—including the weaker Vicuna-based VLMs—emit multiple-choice responses in the correct format essentially without exception. The observed differences in fine-grained accuracy therefore reflect genuinely improved knowledge of the presented options rather than mere gains in format adherence, which isolates knowledge, not answer formatting, as the operative mechanism behind the LLM effect.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most of the evaluated VLMs, the LLaVA series and Molmo among them, adopt OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder, whereas Qwen2-VL (Wang et al., 2024b) employs DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which delivers markedly stronger zero-shot performance on object recognition tasks such as ImageNet. We contrast the two encoders across the full range of LLM choices and pretraining configurations, so that the encoder effect is likewise estimated independently of those axes.

Findings. Figure 5 demonstrates that the benefit of substituting CLIP L/14 with the stronger DFN-CLIP H/14 is contingent on whether connector pretraining precedes finetuning—the encoder helps only once it is properly integrated. Absent pretraining, the mere presence of a stronger encoder is insufficient: DFN-CLIP slightly degrades general VQA performance (-2.5pp) while only modestly improving fine-grained scores (+2.8pp). Once connector pretraining is in place, however, the picture changes decisively: general VQA performance holds steady or improves marginally (+1.2pp), while fine-grained performance rises substantially (+4.5pp). A stronger vision encoder thus translates into a large and sharply targeted gain in fine-grained classification—but only when the encoder is properly integrated into the model through pretraining, making integration, not raw encoder quality alone, the operative condition.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are trained almost universally in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, optimizes a captioning objective on large-scale image-caption data to establish alignment between the vision encoder and the LLM. The second stage, instruction tuning, refines the aligned model on carefully curated multi-round, multimodal instruction-completion data, aligning it with human intent for real-world use. Because these two stages differ sharply in objective, scale, and the parameters they update, we quantify the contribution of each to the fine-grained capabilities of VLMs separately, and find that the two stages play markedly unequal roles.

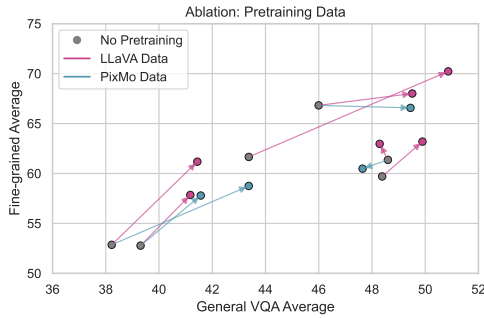


Figure 6: **Pretraining-data ablation.** Relative to no pretraining, adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) connector-pretraining data yields comparable gains: LLaVA data raises fine-grained and general scores by +4.0pp and +2.1pp, and PixMo data by +2.4pp and +2.5pp, respectively. Despite a large disparity in caption quality between the two sources, their effects are nearly interchangeable, indicating that the existence of connector pretraining—rather than the fidelity of its captions—is what drives the gain in this regime.

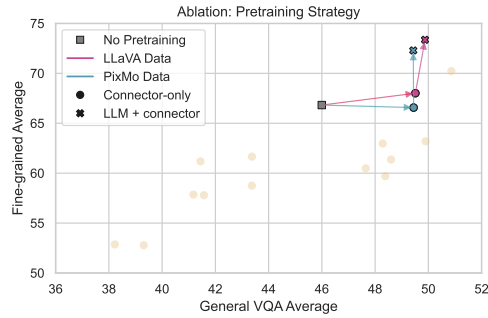


Figure 7: **Training-method ablation: connector-only pretraining versus joint LLM+connector pretraining.** Unfreezing the LLM during pretraining improves fine-grained performance by +5.5pp with no adverse effect on general VQA scores. Because the two schemes differ only in which weights are updated, the gain identifies *which* parameters are trained—not merely *whether* pretraining occurs—as a decisive lever for fine-grained classification.

Table 1: **Summary of the principal ablation settings, with existing models bolded for reference.** Parenthesized values report the change relative to the row above. Upgrading the base LLM and the vision encoder, together with pretraining the connector and LLM on sufficient data, each contributes substantial gains in fine-grained performance, whereas instruction finetuning contributes comparatively little—establishing pretraining and base-model choice, not finetuning, as the principal determinants of fine-grained accuracy. General VQA benchmarks exhibit the complementary pattern, with the choice of LLM accounting for the largest share of the improvement, so that the two capabilities trace to largely distinct components.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

4.3.1 PRETRAINING

Setup. We address three questions that jointly determine how the pretraining stage should be configured:

- Is a dedicated pretraining stage necessary for LLaVA-architecture models, given that their base components are already pretrained on billion-scale data?
- How does connector-only pretraining compare with jointly unfreezing the LLM?
- To what extent does the quality of the pretraining data govern downstream performance?

To resolve these questions jointly, we compare three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, comprising high-quality, human-annotated, detailed cap-

tions), and no pretraining—a design that varies data presence and data quality independently. In most configurations we pretrain only the connector, following Liu et al. (2023). We additionally examine a strategy in the spirit of Deitke et al. (2024), pretraining the connector alone for the first 20% of steps as a warmup before unfreezing the LLM for the remaining 80%, which lets us separate the effect of pretraining from the effect of which weights it updates. Every configuration is followed by one epoch of finetuning on the LLaVA finetuning data, so that only the pretraining stage varies across the comparison.

Findings. Figure 6 reports our pretraining ablation, and the results answer the three questions in turn. Connector-only pretraining improves VLM performance in nearly every configuration, the sole exception being the Qwen2 and CLIP ViT-L/14 pairing, so a dedicated pretraining stage is warranted even though the base components are already pretrained at billion scale. The gains are especially pronounced for Vicuna-based VLMs, which improve by an average of +6.1pp on fine-grained classification against +3.1pp on general VQA—establishing that even lightweight connector pretraining confers a substantial benefit that falls disproportionately on fine-grained recognition.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

More strikingly, the quality of the pretraining captions barely registers: pretraining on low-quality web-scraped captions (LLaVA) and on highly detailed human annotations (PixMo) yields almost identical outcomes, with PixMo differing from LLaVA by only -1.6pp on fine-grained performance and +0.4pp on general VQA—a negligible spread given the order-of-magnitude gap in caption richness.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

We attribute this insensitivity to a specific mechanism: connector-only pretraining leaves the LLM frozen and thereby prevents the model from assimilating the richer signal carried by higher-quality captions, so the additional detail in PixMo has no parameters through which to take effect. This account makes a testable prediction—that unfreezing the LLM should unlock a benefit that connector-only pretraining cannot—which we examine by pretraining the LLM jointly with the connector, prefaced by a 20% connector-only warmup. Figure 7 confirms the prediction: the joint scheme markedly improves fine-grained classification (+5.5pp) while leaving general VQA scores intact. The improvement, moreover, materializes under both LLaVA and PixMo pretraining data, which further establishes that training on the lower-quality captions does not degrade the LLM’s language capabilities.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. Departing from the preceding protocol, we begin from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b)—the model from which Qwen2-VL-7B-Chat is derived—and finetune it on LLaVA-Instruct data (Liu et al., 2023) following our earlier experimental protocol. Comparing the result against Qwen2-VL-7B-Chat, which is instruction-finetuned on 2M in-house examples (Wang et al., 2024b), isolates the effect of the finetuning dataset while holding the base model and pretraining fixed, so that any difference is attributable to instruction tuning alone.

Findings. Substituting the far smaller LLaVA instruction set for Qwen2-VL’s 2M-example corpus lowers both metrics only slightly: fine-grained accuracy falls by 2.4pp (from 87.9% to 85.5%) and general accuracy by 1.3pp (from 62.4% to 61.1%). As Figure 8 makes clear, this is the smallest contribution to fine-grained performance among all our ablations, which establishes that instruction finetuning plays a distinctly secondary role in fine-grained knowledge acquisition relative to base-model selection and pretraining strategy—and that fine-grained knowledge is predominantly acquired earlier, during pretraining, rather than instilled at the instruction-tuning stage.

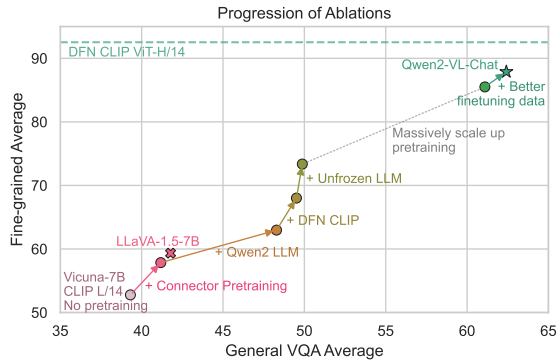


Figure 8: **Cumulative effect of the ablated components.** Beginning from Vicuna-7B with CLIP ViT-L/14, we successively upgrade the base LLM, the vision encoder, and the training settings, raising fine-grained classification accuracy from 52.8% to 73.4%—a 20.6pp gain that closes more than half of the 35.1pp gap to Qwen2-VL-Chat’s 87.9%. The residual difference is attributable primarily to Qwen2-VL’s far larger pretraining corpus of 1.4T tokens, consistent with pretraining scale being the dominant remaining factor.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize our ablations, tracing how each component contributes to closing the gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Starting from the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) with CLIP ViT-L/14 (Radford et al., 2021)), the systematic upgrades we introduce raise fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The decomposition is itself the central contribution of this analysis: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining benefit fine-grained capabilities disproportionately relative to general ability, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) accounts for the majority of the gain in general VQA. The two axes of improvement are therefore driven by largely distinct components—a separability that makes fine-grained and general competence independently addressable through targeted design choices.

Even after these substantial gains, a 12-point gap in fine-grained classification relative to Qwen2-VL-Chat remains unexplained by the components we ablate (Table 1). Two candidate factors could in principle account for it: architectural differences and pretraining data scale.

Our evidence points consistently and specifically to pretraining data scale as the dominant remaining factor rather than architecture: the observational results (Section 3.4) and the training ablations (Section 4.3) both identify pretraining as the decisive lever, and the two candidate explanations differ by orders of magnitude in scale. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) experiments used comparatively small corpora—each fewer than 1M image-caption pairs, or roughly 200M and 400M tokens, respectively—whereas Wang et al. (2024b) pretrain Qwen2-VL on 1.4T tokens, three to four orders of magnitude more. This disparity in pretraining scale is by far the most plausible explanation for the residual gap, and it consolidates the analysis into a single actionable conclusion: extensive pretraining is the critical ingredient for developing VLMs with superior fine-grained classification capabilities.

5 LIMITATIONS & CONCLUSION

Our study carries practical limitations that, in turn, delineate concrete opportunities for future work. Under our computational budget we could compare training only at the <1M data-point scale rather than the billion-scale (B) regime employed by some newer VLMs, so the precise extent of the gains at that scale remains to be measured; the consistent direction of our data-scale evidence, however,

594 indicates that our conclusions should strengthen rather than reverse as pretraining data grows, since
595 scale is exactly the lever our analysis identifies as dominant. Newer work additionally proposes al-
596 ternative pretraining strategies whose interaction with fine-grained visual understanding is a natural
597 next target for the same controlled analysis we develop here.

598 In this work, we systematically evaluate state-of-the-art vision-language models (VLMs) on fine-
599 grained classification benchmarks and establish fine-grained visual classification as a crucial yet un-
600 derexplored dimension of VLM capability—one that current general-purpose benchmarks demon-
601 strably fail to measure. Through a controlled analysis of the principal model components and train-
602 ing paradigms, we decompose vision-centric competence into the specific choices that govern it:
603 base-model selection, vision-encoder integration, and, above all, the scale and design of the pre-
604 training stage. This decomposition yields concrete, actionable strategies for building VLMs with
605 markedly stronger fine-grained visual understanding, and thereby for strengthening their reliability
606 in the real-world, long-tailed scenarios that demand precise visual recognition.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Detailed model-ablation results: aggregate fine-grained classification and general VLM scores.** Each row reports one fully specified configuration—vision encoder, LLM, architecture, pretraining data, finetuning data, and weight-updating scheme—alongside its fine-grained classification accuracy (averaged over the four fine-grained benchmarks) and general VLM accuracy (averaged over the three VQA benchmarks). This table provides the complete evidential basis for the averaged comparisons reported in the main experiments.

A FULL RESULTS

For completeness and reproducibility, we report the full ablation results at three levels of granularity—aggregated performance, per-benchmark general VQA scores, and per-benchmark fine-grained classification scores—in Table 2, Table 3, and Table 4, respectively. Together these tables substantiate every averaged figure cited in the main text and expose the per-benchmark consistency of the reported patterns.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As introduced in Section 3.2, we evaluate VLMs and their base vision encoders on four traditional classification benchmarks, each specified in detail below.

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset organized by the WordNet hierarchy, spanning a broad range of supercategories that each subsume multiple fine-grained subcategories. Its test set comprises 50,000 images distributed over 1,000 classes.

The structure of the ImageNet/WordNet hierarchy, however, leaves many classes too visually distinct from the remainder to admit hard multiple-choice distractors—the “person” supercategory, for instance, contains only three classes, precluding the construction of confusable negatives. To address this, and to make the dataset’s scale tractable, we draw on the human-annotated incorrect model predictions of Recht et al. (2019) to select images for which hard negative answer choices can be constructed. Beginning from 19,056 annotated examples, we distill a pool of 5,457 images covering 928 of the original 1,000 classes. Because this curation leaves the surviving classes unevenly populated, we report mean-per-class accuracy for ImageNet, which weights each class equally regardless of its post-curation frequency.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMM	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Detailed model-ablation results on the individual general VQA benchmarks.** Per-benchmark accuracy on MMBench, MMMU, and MMStar for each ablated configuration, whose mean constitutes the general VLM score reported in Table 2. The per-benchmark breakdown confirms that the aggregate patterns hold consistently across all three VQA benchmarks rather than being driven by any single one.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Detailed model-ablation results on the individual fine-grained classification benchmarks.** Per-benchmark accuracy on ImageNet, Flowers, Pets, and Food for each ablated configuration, whose mean constitutes the fine-grained classification score reported in Table 2. The breakdown demonstrates that the component-level gains hold consistently across all four domains, confirming that the effects reflect general fine-grained competence rather than a single benchmark.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species marked by high intra-class variation and pronounced inter-class similarity, with 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories exhibiting challenging variation in pose, lighting, and occlusion, with 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset of 101 food categories displaying substantial visual diversity within each class, with 25,250 test examples.

Question generation. For every fine-grained classification test set other than ImageNet (treated above), we construct hard negatives by running zero-shot classification on the test examples with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We retain the four highest-ranked incorrect predictions as distractors and shuffle them together with the correct label to form the five-way answer set, so that the negatives are precisely the labels the retrieval model finds most confusable with the image.

Prompt formatting. In our multiple-choice VQA format, the question is held fixed as “What type of object is in this photo?”, and the answer choices are rendered as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because different VLMs are trained under different multiple-choice prompt conventions, we adopt each model’s corresponding format from VLMEvalKit (Duan et al., 2024)—for example, “Answer with the option’s letter from the given choices directly.”—so that no model is disadvantaged by a prompt style mismatched to its training.

For CLIP models we apply the standard zero-shot evaluation procedure, which uses no question format; the class names supplied to the text encoder are instead instantiated in the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the initial observational study of Section 3, we rely directly on the scores reported by the Open-VLM Leaderboard (Duan et al., 2024). By default, that leaderboard reports the average over eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This eight-benchmark aggregate serves as the general-VQA axis in Figure 2 and is used only there.

For every subsequent evaluation—of both existing models and the models we train in Section 4—we restrict attention to a subset of multiple-choice VLM benchmarks that admits a more direct and tightly controlled comparison across configurations. Specifically, we evaluate on three benchmarks:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples spanning a broad range of VLM capabilities partitioned between perception and reasoning. Most examples target coarse-grained perception and visual reasoning; a subset labeled “fine-grained perception” does exist, but its categories (action recognition, attribute recognition, OCR, and the like) occupy a domain distinct from the object-level fine-grained classification datasets we study, and therefore do not overlap with our fine-grained axis.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset spanning a wide distribution of domains and applications. Although the full test set holds 10,500 examples, the field conventionally reports performance on the 900-question validation split, which we adopt for comparability with prior work. MMMU probes general knowledge tied to visual perception across art, science, engineering, medicine, business, and the humanities.

MMStar (Chen et al., 2024b): A dataset of 1,500 human-selected questions, curated to span a diverse set of tasks across both domain coverage and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024), ensuring our scores are directly comparable to previously reported numbers. Each existing model carries a predefined evaluation procedure; for our own trained models we apply the LLaVA evaluation code, consistent with our finetuning exclusively on LLaVA-Instruct data.

C TRAINING

All experiments build on the LLaVA-1.5 training codebase (Liu et al., 2023), and, except where an ablation explicitly dictates otherwise, adhere strictly to the LLaVA architecture and training procedure so that the ablated variable is the only source of variation.

Pretraining. Concretely, we perform one epoch of pretraining on either the LLaVA pretraining data—image-caption pairs drawn from CC-3M—or the Molmo pretraining data (PixMo-Cap); for PixMo-Cap we use only the “caption” field and discard the raw audio transcript. Pretraining uses a batch size of 256. In most runs we update only the randomly initialized MLP connector, at a learning rate of $1e-3$, leaving both the vision encoder and the LLM frozen. When ablating the pretraining strategy, we instead warm up the connector alone for the first 20% of steps and then unfreeze and jointly update the LLM at a learning rate of $2e-5$ for the remaining 80%, so that the two schemes differ only in which parameters the later steps optimize.

Finetuning. Finetuning always consists of one epoch on the LLaVA-1.5 finetuning data—LLaVA-Instruct together with VQA training data drawn from several datasets to support proper benchmarking. Throughout finetuning we hold the LLM learning rate at $2e-5$ with a batch size of 128, and the vision encoder remains frozen across both training phases.

For the Qwen2-VL-architecture ablation, we initialize from Qwen2-VL-7B Base and train under the identical procedure, unfreezing only the LLM.

Compute. All models are trained on 4 nodes of 4 A100 (40GB) GPUs each; a run typically completes within 20 hours, with the precise duration scaling with the CLIP model size.

LLaVA vs. PixMo data. The main text repeatedly invokes the disparity in caption quality between the LLaVA and PixMo pretraining data; we make that disparity concrete here. The LLaVA data derives from CC-3M and consists of web-scraped captions notorious for low quality and weak image relevance. PixMo-Cap, by contrast, is collected by having human annotators record a 60–90 second audio description of each image, which is subsequently consolidated into a coherent, detailed caption (Deitke et al., 2024). The gap is stark even at the surface level: LLaVA captions average 9.8 words, whereas PixMo captions average 169 words—a more than seventeenfold difference in length that the paired examples below illustrate directly.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”